



Daric Power Management Application Notes

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1 Introduction

1.1 Power Scheme

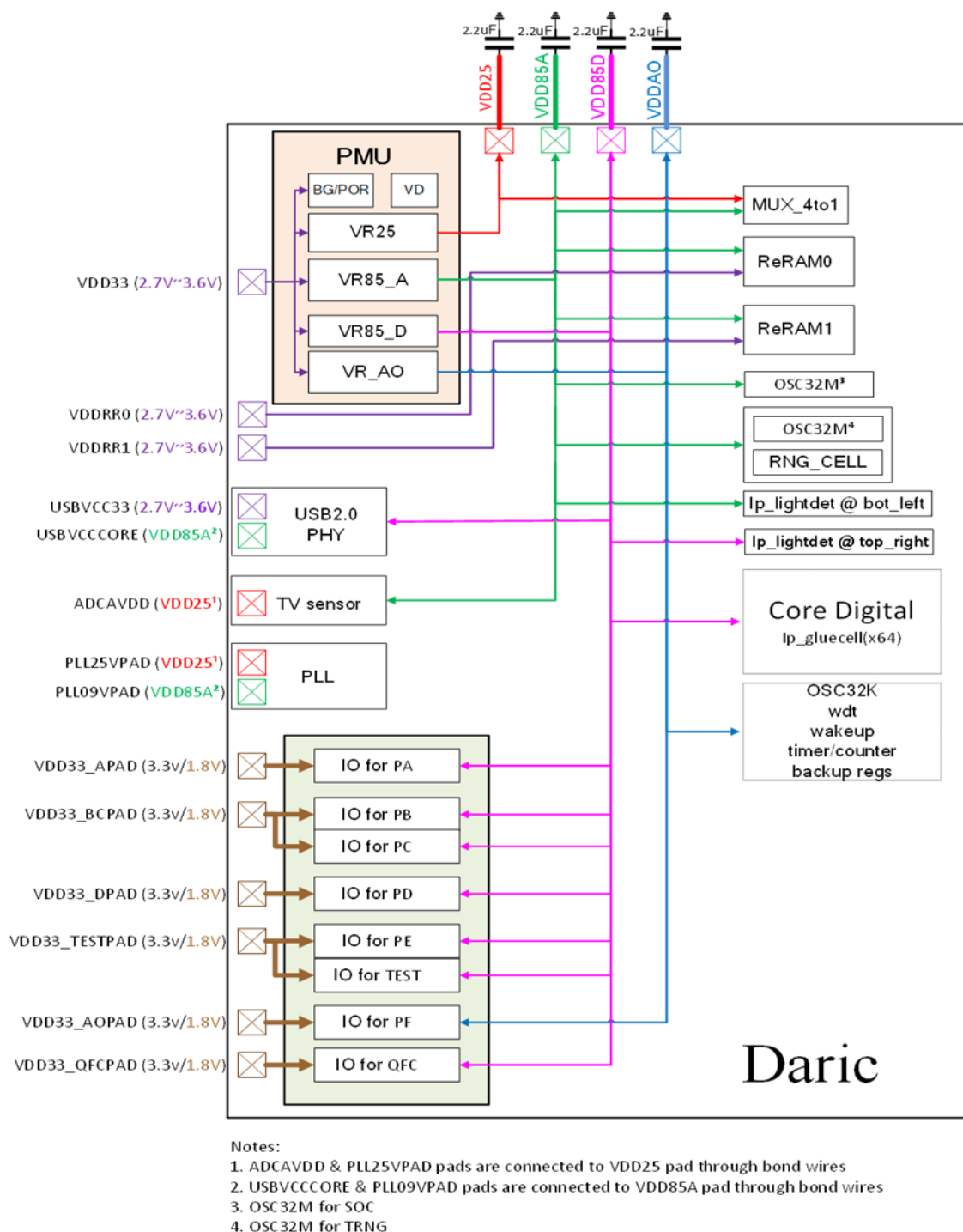


Figure 1-1 Block Diagram

1.2 Power Related Pins

Daric integrates an internal 4-output PMU. All supplies default on when powering-up.

Pin No BGA188	Pin Name	Internal Regulator	Voltage Range (V)	Description
Refer datasheet	VSS	N/A	0V	Ground, analog and digital
	VDD33	N/A	2.7-3.6	Main supply for system and regulators
	VDD85D	Yes, 100mA	Typ 0.8/0.95	Digital supply for CPU, RAM
	VDD85A	Yes,	Typ 0.8/0.95	Analog supply
	AVDD25	Yes,	Typ 2.5	Analog supply
	VDDQFCIO	No	1.62-3.6	QFC IO supply
	VDDRR	No	1.8-3.6	ReRAM supply
	VDDADC	No	Typ. 2.5	Analog supoply for ADC
	PLL25	No	Typ. 2.5	Analog supply for PLL
	PLL09	No	Typ. 0.8	Supply for PLL
	VDDIOA	No	1.6-3.6	Supply for GPIO A bank
	VDDIOBC	No	1.6-3.6	Supply for GPIO B and C bank
	VDDIOD	No	1.6-3.6	Supply for GPIO D bank
	VDDIOE	No	1.6-3.6	Supply for GPIO E bank and system IO's
	VDD33AO	No	1.8-3.6	Supply for GPIO F and Always-On domain
	VDD08AO	Yes	Typ. 0.8	Supply for RTC and always-on RAM
	USBVDD33	No	3.0-3.6	Supply for USB IO
	USBVCCCORE	No	Typ. 0.8	Supply for USB digital

Table 1-1 Power Related Pins

1.3 Power-Up Sequence

Pin Name	Internal Regulator	Power-up Sequence	Description
VDD33	N/A	2	Mandatory supply
VDD85D	Yes, 100mA	1	Mandatory supply
VDD85A	Yes,	3 (internal VR by default)	Mandatory supply
AVDD25	Yes,	3 (internal VR by default)	Mandatory supply
VDDQFCIO	No	4	Optional, as needed
VDDRR	No	1	Mandatory supply
VDDADC	No	3 (Internal VR or external)	Optional, as needed
PLL25	No	4	Optional, as needed
PLL09	No	4	Optional, as needed
VDDIOA	No	4	Optional, as needed
VDDIOBC	No	4	Optional, as needed
VDDIOD	No	4	Optional, as needed
VDDIOE	No	1	Mandatory supply
VDD33AO	No	0 (Always-On)	Mandatory supply
VDD08AO	Yes	0 (external) / 3 (internal VR)	Mandatory for AO RAM and AO registers
USBVDD33	No	4	Optional
USBVCCCORE	No	4	Optional

Table 1-2 Power-up Sequence

2 Power Modes

2.1 Power Modes

Supplies	Active	Sleep	Deep Slp	PD with PMU	Data retention PD	Power Off
VDD33	On	On	On	On	Off	Off
VDD85A/D	Int/Ext	Int/Ext	Int/Ext	Off	Off	Off
ReRAM	On	On	On	Off	Off	Off
GPIOE	On	On	On	Off	Off	Off
Other power	On	On	On	Off	Off	Off
V33AO	On	On	On	On	On	On/Off
V08AO	Int/Ext	Int/Ext	Int/Ext	Int/Ext	Ext	Off
Typical System consumption	30-300 mW	Slightly lower than Active	40 mW	2-4 mW	< 1 mW	<300uW (PMIC)
SRAM Detention	Yes	Yes	Yes	No	No	No
AORAM Retention	Yes	Yes	Yes	Yes	Yes	No
Enter condition	Boot/int/evt	WFI/WFE	WFI/WFE			
Wake-up source	N/A	Int/Evt #3	Int/Evt #3	AOIO Int#4, #5	Power-up #5	Power-up #5
Wake-up latency	N/A	~ 100ns	us-ms(PLL re-config)	sec	sec	sec
Known Issues	-	-	-	DCDC start #1	-	PortF status #2

Table 2-3 Power Modes

Notes:

1. Some PMIC's/Regulators might fail to start when VDD85A/D generated by PMU
2. PortF output/Pull-up reset after VDD33 off.
3. Any Interrupt/Event
4. AO events, PF0/1
5. Reboot

2.2 Performance Modes

TBD

3 Design Recommendations

3.1 Absolutely Minimal System

For minimum BOM, SIP, even Smartcard

Power Rail	Supply Source	Comments
VDD33	External	Main supply
VDD85D	Internal PMU	
VDD85A	Internal PMU	
AVDD25	Internal PMU	
VDDQFCIO	N/A	Likely not used
VDDRR	Tie to VDD33	
VDDADC	Internal PMU, AVDD25	
PLL25	Internal PMU, tie to AVDD25	Likely not used, running at fixed frequency or lower
PLL09	Internal PMU, tie to VDD85A	Likely not used, running at fixed frequency or lower
VDDIOA	Tie to VDD33	As needed
VDDIOBC	Tie to VDD33	As needed
VDDIOD	Tie to VDD33	As needed
VDDIOE	Tie to VDD33	
VDD33AO	Tie to VDD33	
VDD08AO	Internal PMU	
USBVDD33	Tie to VDD33	
USBVCCCORE	Tie to VDD85D	

Table 3-4 Minimum BOM

3.2 Typical Battery-Powered

Most common case, external DCDC regulators recommended. PMIC preferred.

Power Rail	Supply Source	Comments
VDD33	External	Main supply
VDD85D	External DCDC	For efficiency. DVFS highly recommended
VDD85A	Internal PMU	Very low consumption
AVDD25	Internal PMU	Very low consumption, linear VR will be ok
VDDQFCIO	External DCDC	As needed
VDDRR	External DCDC	1.8-3.0 V for lowest power consumption
VDDADC	Internal PMU, AVDD25	Linear VR will be OK
PLL25	Internal PMU, tie to AVDD25	Very low consumption, linear VR will be ok
PLL09	External Low-noise	As needed, Approx. 2mA
VDDIOA	External 1.62-3.6 V	As needed
VDDIOBC	External 1.62-3.6 V	As needed
VDDIOD	External 1.62-3.6 V	As needed
VDDIOE	External 1.62-3.6 V	
VDD33AO	1.8-3.3V	Always On. Or Controlled for lowest power off consumption
VDD08AO	Internal PMU	Always On. Or Controlled for lowest power off consumption
USBVDD33	External	As needed
USBVCCCORE	External	As needed, Approx. 2mA

Table 3-5 Typical battery-powered

4 FAQ's

TBD

5 Design Review Information

Topic of Review:

Date:

Related Documents:

Meeting Moderator:

Meeting Scribe:

Invitees	Attended (Y/N)	Invitees	Attended (Y/N)

Agenda/Issues discussed:

Outcome/Summary:

ID	Item	Owner	Date Due	Status	Closed?
1					
2					
3					

Table 5-6 Actions Items